SCHOOL OF COMPUTER SCIENCE

APPLIED MACHINE LEARNING PROJECT

PREDICTING THE RESULTS OF PROFESSIONAL TENNIS MATCHES WITH MACHINE LEARNING

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Declaration

I hereby declare that the work described in this report is, except where otherwise stated, entirely my own and that I have not submitted it before for summative assessment in third-level education. I have read and understand TU Dublin's policy on plagiarism.

Signed .... Joe Bloggs ....   Date ...... 02/04/2026 ......

(Sergio de Carvalho, D24130831)

Acknowledgements

[Please leave this section out if you do not have any acknowledgements.]

Abstract

[This should be up to 200 words and provide a summary of what was done and important findings.]

Contents

Acronyms and Abbreviations

ANP	Analytic Network Process
ATP	Association of Tennis Professionals
DT	Decision Tree
EDA	Exploratory Data Analysis
GBM	Gradient Boosting Machine
LASSO	Least Absolute Shrinkage and Selection Operator
LGBM	Light Gradient Boosting Machine
ML	Machine Learning
MLP	Multi-Layer Perceptron
RF	Random Forest
SVM	Support Vector Machine
WTA	Women's Tennis Association
XGBoost	eXtreme Gradient Boosting

1 Introduction

Tennis is a popular sport where the one-on-one nature of competition provides an ideal environment for sports analytics, as match outcomes are determined by a more controlled set of variables than in team-based sports. While professional tennis has benefited from rigorous record-keeping and a wealth of statistics about every aspect of the game, accurately predicting match results remains a complex challenge due to the intricate interactions between player skills, form, physical and psychological state, as well as environmental factors such as the weather, court surface type, and even tennis ball characteristics that can significantly influence the performance of players and the outcome of a match.

1.1 Project description

The focus of this project is to develop a framework that prioritises the curation of high-quality data and the engineering of domain-driven features that can capture the multiple facets of the game and be used to predict the results of professional tennis matches. Central to this approach is the use of "enhanced covariates"—variables that are not directly observed but are derived through mathematical transformations or separate statistical models.

There has been significant research on developing statistical models to rank tennis players as alternatives to the official rankings published by the professional tennis organisations. In parallel, several studies have applied machine learning to predict the results of tennis matches; however, most of these studies were restricted to specific tournaments (e.g. Grand Slam tournaments only) or did not take full advantage of the available data and statistical models. This project aims to address these limitations by tapping into the vast amount of data available and using machine learning techniques to explore the predictive power of different statistical models proposed in the literature while also discovering new ways to improve the predictions of these models.

1.2 Project motivation and justification

Predicting the outcomes of sporting events has been a long-standing problem in the field of sports analytics, and can have significant implications for the development of training methods and strategies to improve the performance of professional athletes. It is also important for broadcasters and the media in general as it helps them to transform a passive viewing experience into an interactive, high-stakes narrative that drives viewer retention and fan engagement. Finally, it can be argued that predicting sporting events can help develop valuable models and gain insights that could be useful in diverse areas of human activity, from financial markets, to politics and international conflicts (Gu and Saaty 2019).

1.3 Project scope

This project is focused on predicting the results of professional tennis matches at tournaments on the men's tour organised by the Association of Tennis Professionals (ATP). While there is a wealth of data available from the Women's Tennis Association (WTA) as well, limiting the scope to one of the two circuits allows for a more focused and comprehensive analysis in the limited time available for this project. We note that the focus of this project is modelling predictions based solely on pre-match data rather than using data collected while the match is in progress.

Finally, we note that there is significant interest in the literature in using betting odds to either improve predictions or compare their accuracy. While odds that are carefully curated and published by professional bookmakers can be a valuable source of information, this project does not use odds data nor does it focus on this aspect of the problem.

1.4 Report layout

[A sentence on each of the following sections, introducing the reader to the report.]

2 Literature review

The Association of Tennis Professionals (ATP) manages the men's circuit and maintains the official player ranking that is used to determine tournament eligibility, entry qualification, and player seeding. This ranking is based on a cumulative rolling points system. Players earn points for each match won in an official tournament, and points remain on a player's record for 52 weeks before expiring. The number of points awarded for each win is determined by the event category and the round in which the match is played (Williams et al. 2021). For instance, the champion of any of the four annual Grand Slam events (Australian Open, Roland Garros, Wimbledon, and the US Open) receives 2,000 points, whereas the winner of a first-round match in the same tournament who does not advance any further receives 50 points. In contrast, the champion of an ATP 250 event receives only 250 points.

While official rankings are a good indication of the current strengths and abilities of tennis players, and can generally be used to predict the winner of a professional tennis match, they obviously do not capture the full story and are far from being a reliable prediction tool, particularly when the difference in rankings between players is small.

2.1 Paired comparisons and rating mechanisms

Noting that official systems often reward frequent participation over absolute performance, researchers set out to develop alternative rating mechanisms that could capture players' abilities by looking at their past match results, particularly taking into account their opponents' strengths rather than the prestige of the tournament or the round in which the match is played.

One of the earliest attempts to develop an improved rating mechanism for tennis was proposed by Glickman (1999), who devised a dynamic paired comparison model to estimate the abilities of players over time. He showed that, given a particular choice of constants, his method is equivalent to the popular Elo rating system developed for chess players by Arpad Elo in 1978. While Glickman used match results to produce an alternative ranking for tennis players, he did not test its forecasting capabilities.

McHale and Morton (2011) pioneered a similar rating mechanism by building a Bradley-Terry type model (a popular model for handling paired comparisons) that utilised players' past match results using an exponential decay function to account for players' recent form (i.e. giving more importance to recent results than older ones). Their model also accounted for the score of the matches. For instance, a player who wins a match with a score of 6-0 6-0 is rewarded more than a player who wins with a score of 7-6 7-6 (presumably, a much tighter match). They also note that tennis is played on different surfaces (e.g. clay, grass, and hard courts) that can significantly change the performance of players, thus they also made further use of weighting techniques so that matches played on surfaces other than that of the tournament being considered are given less importance. They showed that their model provided superior forecasts to the official rankings using data from nine seasons of the ATP tour from 2000-2008, and could provide higher returns against bookmaker odds using a simple betting strategy.

Expanding on this relational logic, Knottenbelt, Spanias, and Madurska (2012) proposed a common-opponent stochastic model that relied on the transitive nature of tennis, predicting match outcomes by calculating the difference in service points won against opponents that both players had previously faced. They argued that the intuition behind this approach is that tennis is a transitive sport to a certain extent (i.e. if player A is better than player B and player B is better than player C, then, usually, player A is better than player C). However, as many avid tennis fans will attest, this is often not the case. Using this approach, they were nevertheless able to generate 3.8% long-term profit against the best odds offered by bookmakers for a set of over 2,000 tennis matches from the ATP and WTA tours of 2011.

Kovalchik (2016) tested the predictive performance of 11 published forecasting models for predicting the outcomes of 2,395 singles matches during the 2014 season. They found that an Elo-based model was the best performing model, achieving an accuracy of up to 70%, although this was still worse than a bookmaker consensus model built by aggregating the odds offered by established bookmakers for each match, which achieved an accuracy of 72%.

Subsequent research heavily explored the Elo rating system. Williams et al. (2021) compared standard Elo, surface-specific Elo ratings, and weighted composites against betting odds, demonstrating that Elo-based approaches are effective, offering comparable, and sometimes superior, forecasting performance to betting odds on Grand Slam matches between 2018 and 2020. Angelini, Candila, and De Angelis (2022) introduced the concept of a Weighted Elo (WElo), which incorporates the margin of victory (i.e., the number of games or sets won) into the standard Elo updating formula to better reflect players' strengths. They tested their model on a set of matches from the ATP and WTA tours between 2012 and 2020, and showed that the WElo was superior to the standard Elo approach as well as the Bradley-Terry type model of McHale and Morton (2011), among other models.

Fayomi et al. (2022) revisited the Bradley-Terry model, using it to compute win probabilities and new player ratings that demonstrated the profound impact of court surface (especially, clay vs. hard court) on a player's true ability. They applied the model to data from the ATP tour from January 2019 to September 2020, achieving results similar to those of McHale and Morton (2011).

2.2 Machine learning approaches

Predictive techniques have transitioned from statistical models to complex machine learning (ML) approaches. An early example is Somboonphokkaphan, Phimoltares, and Lursinsap (2009), who deployed a Multi-Layer Perceptron (MLP) – a foundational class of fully-connected feedforward artificial neural network – that incorporated time-series history, environmental data in the form of surface type, and several statistical features such as winning percentage of first serves and total points won. They tested their model on a set of matches between 2003 and 2008 and reported accuracy rates ranging from 69% to 81%, although it should be noted that these results were restricted to a small set of Grand Slam matches only.

Recognising that historical data alone misses intangible factors, Gu and Saaty (2019) built an Analytic Network Process (ANP) – an advanced, non-linear decision-making framework – combining data analysis with expert human judgments, capturing unquantifiable variables such as a player's psychology, brainpower, and stamina. They reported that their model achieved an accuracy of up to 85.1%; however, once again, this was restricted to a set of only 94 matches from the 2015 US Open tournament. Moreover, this approach required the collection of judgements from tennis experts based on their knowledge and expertise about the matches and the players, an approach that is clearly difficult to scale.

Gao and Kowalczyk (2021) built three machine learning models (Logistic Regression, Random Forest, and Support Vector Machine), performed an in-depth feature selection process, and compiled one of the largest tennis statistics datasets publicly available. They found that a Random Forest classifier achieved the best performance, achieving prediction accuracies exceeding 80%. They also identified serve strength—specifically the percentage of points won on first and second serves—as a key predictor of match outcomes. Wilkens (2021) conducted a comprehensive comparative study using various ML methods (Logistic Regression, Neural Network, Random Forest, Gradient Boosting Machine, and Support Vector Machine) on a dataset of over 39,000 matches from the ATP and WTA tours, and noted that differences in prediction performance among the various ML approaches were small.

Yue et al. (2022) applied exploratory data analysis (EDA) to examine variables related to match results and introduced new variables using the Glicko rating system of Glickman (1999), including surface-specific ratings, which were then used to build several statistical and machine learning models (Logistic Regression, Support Vector Machine, Neural Network, and Light Gradient Boosting Machine). The models were trained on a dataset of Grand Slam matches between 2000 and 2019, and the results showed only marginal differences in performance between the various models, with the best results achieved using surface-specific Glicko ratings, with up to 73.8% accuracy on the ATP matches.

Buhamra, Groll, and Gerharz (2025) compared several models including regression-based (logistic, LASSO, splines) and machine learning approaches (Classification Tree, Random Forest, linear and non-linear Support Vector Machine, Artificial Neural Network, and XGBoost) using Grand Slam matches from 2011 to 2021. Different features were considered including ATP ranking and points, Elo ratings, bookmaker odds, and age variables that take into account the presumably optimal age of a tennis player being between 28 and 32 years old. Moreover, they investigated different forecasting strategies including “expanding window” and “rolling window”. They found that XGBoost achieved the highest classification rates, with up to 83% accuracy with the expanding window strategy.

2.3 Identified research needs

Taken together, the studies reviewed in this section illustrate a recurring limitation: conclusions are often drawn from short calendar windows or focus almost exclusively on Grand Slam tournaments. Moreover, as noted by Wilkens (2021), the heterogeneous setup and data used throughout the literature also demand prudence when comparing or even generalising their findings. These shortcomings motivate the emphasis for the present work: assembling and curating a large, multi-decade body of high-quality professional tennis match data, enriching it with derived metrics and several alternative rating formulations (including variants inspired by Elo- and Glicko-type ideas), and, finally, applying multiple machine learning approaches and comparing them under a common protocol, so that differences in predictive performance can be interpreted in light of consistent data and feature engineering.

3 Methodology

3.1 Data collection

Historical ATP data was obtained from an open repository maintained by Jeff Sackmann[1], which contains comma-separated (CSV) files of ATP players, rankings and match results. This repository was cloned into a separate fork[2] so that a small number of data-quality issues discovered during experimentation could be corrected in a controlled (and documented) manner. Examples of such fixes include removing duplicate ranking rows, excluding matches in which the placeholder name "Bye" appeared as if it was a player, as well as a few other minor inconsistencies in the source files.

The repository presents information in three main families of CSV files:

(1) **Player files** supply a unique numeric player identifier (ID) together with the player name, playing hand (right, left, ambidextrous, or unknown), date of birth, and height.

(2) **Ranking files** contain the official ATP weekly rankings: ranking date, player ID, ranking position, and ranking points.

(3) **Match files** describe individual completed contests: tournament information (name, level, starting date, court surface, draw size), player IDs for the winner and loser, match format (e.g. best-of-three or best-of-five sets), and the recorded scoreline.

Match files were combined into a single match-level table. Each row was then linked to the player master file on winner and loser IDs so that player attributes could be attached to each match. Ranking files were aligned to matches by selecting, for each player, the ranking position and points at the tournament start date (or the closest available ranking date). Winner and loser player IDs were randomly reassigned as player 1 and player 2, so that the dataset is not biased towards one player or the other. A winner column was added to indicate the winner of the match. The consolidated dataset resulted in 189,373 ATP singles matches spanning the seasons from December 1967 through December 2024. Figure 1 shows the distribution of matches per year by tournament level (Grand Slam, Masters 1000, ATP Finals, etc.).

3.2 Feature engineering

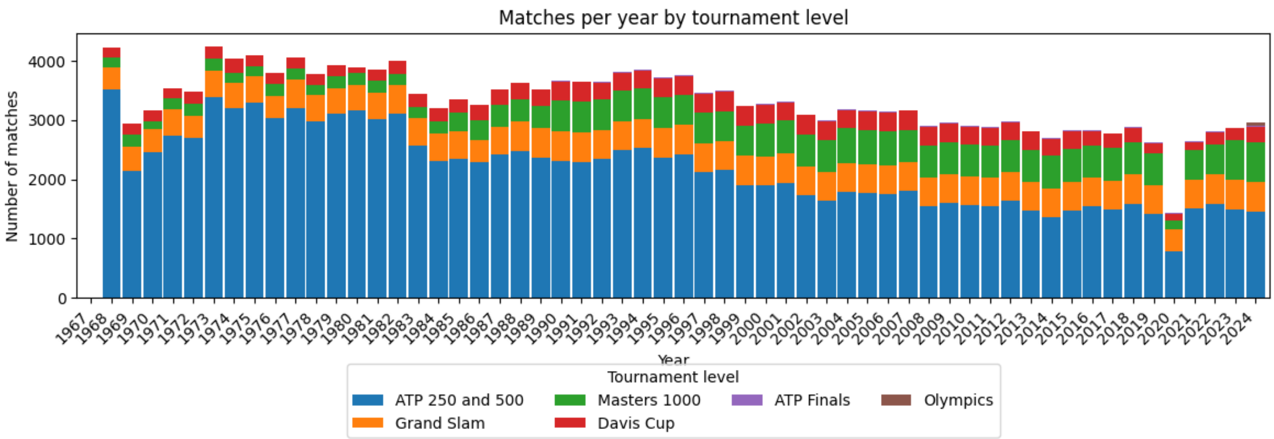
Several features were calculated in a single chronological pass over all matches, with a running state maintained for every player (and for every player–surface pair) so that quantities could be computed from *prior* contests only. In other words, match outcomes were written to the exported row first, and only afterwards were win/loss counters, head-to-head tallies, and rating updates advanced to include the current result. This guarantees that no statistic leaks information from matches played later in time or from the fixture being predicted.

For each side in a match, the following history-based quantities were derived from official ATP ranking snapshots and from completed matches strictly before the match under consideration:

[1] `https://github.com/JeffSackmann/tennis_atp`

[2] `https://github.com/scarvalhojr/tennis_atp`

Figure 1: Distribution of matches per year by tournament level



- **Career-best ranking.** The best (numerically smallest) official singles ranking position the player had attained up to the tournament week.
- **Career match volume and win rate.** Total professional matches played to date and the fraction of those matches won.
- **Recent activity and form.** The number of matches played in approximately the last twelve months and the fraction of those matches won. This "year" window is a rolling look-back of 365 days from the tournament date (rather than calendar months).

Because surface strongly conditions playing style and results, parallel counters were maintained for each court type on the ATP tour (hard, clay, grass, and carpet). For every player and surface, the same career totals and twelve-month counts and win percentages were tracked using only matches played on that surface before the current match. Where the surface was unknown or missing, surface-specific aggregates were left undefined for that row.

Head-to-head counts were also attached to each match, recording how many times the first-named player had beaten the second to that date, in two directed views: overall (i.e. all surfaces) and on the current event surface. Symmetric statistics for the opponent were also added.

Alongside the tabular features above, the same chronological scan maintained eight parallel rating histories for every player: four derived from Elo-style updates and four from Glicko-style updates. In each case the variants are (i) overall versus surface-specific, and (ii) binary win–loss versus *margin-of-victory*-weighted updates. The surface-specific variants are updated only from prior matches played on the same surface, whereas the overall variants are updated from all prior matches regardless of surface. The weighted variants use a margin-of-victory factor V in the spirit of McHale and Morton (2011): $V = G_{\text{win}}/(G_{\text{win}} + G_{\text{lose}})$, where G_{win} and G_{lose} are total games won in the match by the winner and loser, respectively (from the recorded scoreline, independent of which player is listed first). For instance, a 6–0 6–0 score yields $V = 1$, whereas a 7–6 7–6 result yields $V \approx 0.538$.

Elo-type ratings. Every player begins from the conventional initial rating of 1500. After each completed match, the rating change follows the usual logistic expected-score pairing based on a 400-point rating scale; the step size is damped as the player's career match count grows, using the scaling factor $K = 250/(N+5)^{0.4}$, where N is the player's match count prior to the match under consideration, as proposed by Kovalchik (2016).

Glicko-type ratings. Glicko ratings use the Glicko-2 variant documented in Glickman (2022), with non-overlapping *rating periods* of 60 days and system volatility parameter $\tau = 0.5$. In this system, each player has a rating, a rating deviation, and a volatility. The rating deviation measures the uncertainty of the player's rating, and is high when a player is not competing frequently. The volatility indicates the degree of expected fluctuation in a player's rating — it is high when a player has erratic performances and low when the player performs at a consistent level. Matches within a rating period are treated as having occurred simultaneously. The margin-weighted variant maps the games ratio V to a Glicko-2 outcome factor $M = 0.5\,V + 0.5$, since the algorithm expects a score between 0 and 1, where 0 means the first player in the pairing lost, 0.5 means a draw, and 1 means the first player won.

Together, each entrant therefore carries four Elo-based features (`elo`, `welo`, `surface_elo`, `surface_welo`) and four Glicko-based features (`glicko`, `wglicko`, `surface_glicko`, `surface_wglicko`) prior to the match being predicted.

Table 1 in Appendix A shows the full set of features with brief descriptions.

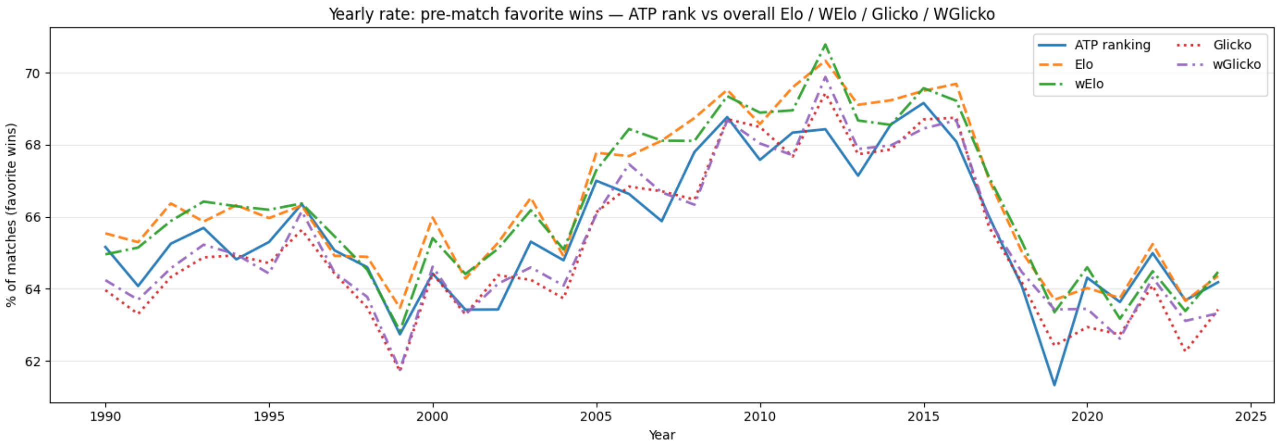
3.3 Data analysis

The chronological feature pipeline described above was run over the full consolidated match history so that counters, head-to-head totals, and rating systems (including all Elo and Glicko variants) reflect every available contest in temporal order. When turning to supervised machine learning, however, training and evaluation were restricted to rows where *both* entrants had an official ranking position and ranking points available at the time of the match. Matches missing either value for either player were excluded because exploratory analysis suggested that ranking-based inputs were among the strongest predictors of the winner, and retaining rows with missing rankings would either weaken those signals or require ad hoc imputation. In practice, this filter removes essentially all matches before 1990, when systematic ranking snapshots for both opponents become sufficiently complete in the source data. The resulting dataset contains 104,092 rows.

That emphasis on rankings is supported by a simple sanity check: among matches where both players have known rankings, define the *favorite* as the player with the better (numerically lower) ranking position; that player's win rate is far above 50% in every season of the dataset. In fact, the same idea applies to the Elo and Glicko ratings (including their margin-weighted variants): treating the higher-rated player before the match as the favorite also yields win percentages well above chance and, hence, these inputs remain powerful candidates for supervised models.

Figure 2 summarises this behaviour year by year on the filtered data subset used for modelling (both ranks and both ranking points present for each player). Each series plots the percentage of matches in which the pre-match favorite—by ATP ranking, Elo, weighted Elo (wElo), Glicko, or weighted Glicko (wGlicko)—won the contest. All lines sit clearly above a naive 50% baseline and generally track one another. Surface-specific ratings are not shown but their predictive power is similar.

Figure 2: Yearly share of matches won by the pre-match favorite by ATP ranking (lower is better) versus overall Elo, weighted Elo (wElo), Glicko, and weighted Glicko (wGlicko); matches where both players have known rank and ranking points only.



3.4 Data preparation

Beyond excluding rows with unknown ranking position or ranking points for either player (as set out above), the modelling dataset required little further preparation. A handful of remaining fields still contained occasional missing values; these were addressed with straightforward imputation in the machine-learning pipelines using median or most frequent values.

For supervised experiments, columns were grouped by role. The **target** variable was the match `winner` indicator (1 or 2). **Metadata** columns—carried for traceability but not fed as inputs to the classifiers—comprised of tournament information (ID, name, start date), match order within the event, the recorded score string, and both players' identifiers and names. **Categorical** inputs comprised match format (`best_of`), tournament level, court surface, and each player's dominant hand; every other predictive column was treated as **numeric** (including rankings, rating histories, head-to-head totals, and derived counts). Categorical features were encoded using one-hot encoding.

The cleaned table was divided into **training** and **testing** subsets with an $80\,\%$–$20\,\%$ row split, respectively. The training pool was then reordered by match date so that a *chronological* 10-fold cross-validation could be used to tune hyperparameters without leaking information from later matches into earlier ones. After selecting hyperparameters on those folds, each model was refit on the full training set with the best hyperparameters. The held-out $20\,\%$ test was then used to report general performance.

3.5 Supervised Machine Learning Models

Three off-the-shelf classifiers were chosen to span complementary inductive biases while staying close to models that appear repeatedly in tennis forecasting work. **Random Forest** provides a strong, interpretable baseline. **XGBoost** implements regularised gradient boosting with second-order optimisation; it often reaches state-of-the-art accuracy on structured data. **CatBoost** rounds out the trio by targeting high-cardinality and categorical fields with ordered target statistics and symmetric trees, reducing sensitivity to the exact one-hot encoding scheme of categorical features. Together, the three families give a concise comparison of mainstream toolkits under the same preprocessing, temporal cross-validation, and hyperparameter search pipeline.

3.6 Implementation details

The processing of the original CSV files into the final match-level dataset with all features discussed in the previous sections was implemented in Python (code is available online[3]). Elo-style ratings are implemented directly within the codebase. Glicko-2 ratings delegate the update mathematics and period processing to a third-party library[4]. Supervised machine learning models were implemented using the scikit-learn framework[5], with the built-in Random Forest classifier, and third-party XGBoost[6] and CatBoost[7] libraries.

[3] https://github.com/scarvalhojr/tennispredictor/tree/main/code/databuilder
[4] https://pypi.org/project/glicko2-py/
[5] https://scikit-learn.org/
[6] https://xgboost.ai/
[7] https://catboost.ai/

4 Results

[You may structure this as you wish, but let the reader know everything you know about the results, regardless of whether you consider the outcome good or bad.

Discuss the results, rather than simply presenting them to your reader and letting them do the work.

Be inquisitive when analysing your results. For example, if you are getting accuracies in the high 90s there is probably either an extreme imbalance in the data set or a data leak. Always investigate and at least discuss but, if there is time, it is best to redesign and redo your experiments.

Discuss the implication of the results in the application domain]

5 Conclusion

[Structure as you wish but make sure that a it includes a synthesis of the results, a high-level discussion of their implications in the application domain, any particularly interesting low-level details of the results and, finally, some words on future work i.e. how you or someone else might build on the project.]

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A Final dataset features

The exported match-level file contains 63 columns. Table 1 lists the full schema with brief descriptions. Features denoted by the placeholder `playerX` stand for *two* CSV columns in each row—one prefixed with `player1` and one with `player2`—carrying the same information for each entrant. All other features appear once per row.

Table 1: Columns in the final dataset CSV.

Feature name	**Description**
Tournament and match metadata	
`year`	Calendar year of the tournament start date.
`tournament_id`	Unique identifier for the tournament instance.
`tournament_start_date`	Scheduled first day of the event (in `YYYY-MM-DD` format).
`tournament_name`	Official tournament name from the source files.
`tournament_level`	Tournament level (e.g. Grand Slam, Masters 1000, etc.).
`surface`	Court surface (e.g. hard, clay, grass, carpet) when known.
`draw_size`	Main-draw size (e.g. 32, 64, 128 players).
`match_num`	Order of the match within the tournament (source ordering).
`best_of`	Match format (e.g. best-of-3 or best-of-5 sets).
Per-player fields (replace `X` with `1` and `2`)	
`playerX_id`	Unique player identifier.
`playerX_name`	Player full name.
`playerX_hand`	Playing hand (right, left, ambidextrous, or unknown).
`playerX_height_cm`	Height in centimetres when available.
`playerX_age`	Age in years at the tournament start date.
`playerX_rank`	Ranking position on the ranking date aligned to the match.
`playerX_points`	Ranking points on the ranking date aligned to the match.
`playerX_highest_rank`	Best ranking position achieved up to the tournament week.
`playerX_career_matches`	Count of prior completed professional matches.
`playerX_career_win_rate`	Fraction of prior professional matches won.
`playerX_year_matches`	Count of matches in the 365 days before the match.
`playerX_year_win_rate`	Fraction of matches won in the 365 days before the match.
`playerX_surface_career_matches`	Prior matches on the same surface type as the event.
`playerX_surface_career_win_rate`	Career win rate on that surface (prior matches only).
`playerX_surface_year_matches`	Matches on that surface in the 365 days before the match.
`playerX_surface_year_win_rate`	Win rate on that surface in the 365 days before the match.
`playerX_head2head_wins`	Prior wins by this player against the opponent (directed count).
`playerX_head2head_surface_wins`	Prior wins against the opponent on the current match surface.
`playerX_elo`	Overall Elo-style rating before the match.
`playerX_welo`	Margin-of-victory weighted Elo-style rating before the match.
`playerX_surface_elo`	Surface-specific Elo-style rating before the match.

Continued on next page

Table 1 — *continued*

Feature name	Description
`playerX_surface_welo`	Surface-specific margin-of-victory weighted Elo-style rating before the match.
`playerX_glicko`	Glicko-style rating before the match.
`playerX_wglicko`	Margin-of-victory weighted Glicko-style rating before the match.
`playerX_surface_glicko`	Surface-specific Glicko-style rating before the match.
`playerX_surface_wglicko`	Surface-specific margin-of-victory weighted Glicko-style rating before the match.
Outcome	
`score`	Match scoreline string (sets and tie-break detail).
`winner`	Indicator of which listed player won (`1` or `2`).

B Examples of figure, table and equation

Figure example

Example of figure:

Figure 3: Example caption

And here is an example of a reference to Figure 3.

Table example

Example of table:

Table 2: Correlation values for N=100 and $\alpha = .05$

Pair	r	Significant
First	0.22	Yes
Second	0.07	No
Third	0.31	Yes

An example of referencing Table 2.

Equation example

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{(n-1)S_x S_y} = \frac{S_{xy}^2}{S_x S_y} \tag{1}$$

where x, y are the two variables, x_i, y_i the ith values, $\bar{x}$, $\bar{y}$ the means, S_x and S_y the sample standard deviations and S^2_{xy} the sample covariance. And here is a reference to Equation 1.